

# Georges Birangou

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Casablanca, Morocco

## PROFILE

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Data & AI Engineer with hands-on experience in ETL pipeline design, cloud data migration, and machine learning systems. Skilled in Python, Spark, Kafka, and Azure, I have a track record of delivering end-to-end data solutions, from raw ingestion to production-ready insights, while automating complex workflows. Passionate about building scalable, impact-driven architectures at the intersection of Big Data and AI.

## PROFESSIONAL EXPERIENCE

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### Data & AI Engineer Intern

Mar. 2026 – Present

*Cetim Maroc Développement*

*Casablanca, Morocco*

- Designed and developed a Python system automating HCF test report generation for aeronautical clients (Safran, Aubert & Duval), ensuring full result traceability and eliminating manual data entry errors
- Designed a multi-source ETL pipeline parsing heterogeneous fatigue test files (machine exports: .xlsx, .txt) and data from **Azure Databricks**, applying domain-specific transformations and loading into structured objects for automated report generation, **reducing production time by 35%**
- Integrated a local AI model into the system to assist with automated comment drafting
- PyQt6 desktop interface with real-time PDF preview, packaged as a standalone EXE via PyInstaller for dependency-free deployment

### Software Engineer Intern

Jun. 2025 – Sep. 2025

*Chambre de Commerce de Casablanca-Settat*

*Casablanca, Morocco*

- Designed the PostgreSQL/Supabase database schema and developed a cross-platform library management system (React Native + Next.js) managing **500+** books via **REST APIs**, with real-time synchronization and automated notifications
- Built an analytics dashboard powered by aggregated SQL queries (usage trends, peak activity, reservation volumes), **saving 1.5 hours/week** of manual reporting for library staff

## PROJECTS

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### Real-Time Banking Fraud Detection System | *Python, XGBoost, Kafka, PostgreSQL, Grafana, Docker* 2025

- Trained an XGBoost + SMOTE model on **284,807** transactions to detect fraud in a highly imbalanced dataset (**0.17%** fraud rate), achieving ROC-AUC of **0.978**
- Reached **89.8%** recall on the test set, intercepting **88 out of 98** fraudulent transactions while keeping false alerts at **0.3%**
- Built a real-time Kafka streaming pipeline for transaction ingestion and ML inference, with results stored in PostgreSQL and monitored via a Grafana supervision dashboard
- Deployed the full architecture in Docker for containerized, reproducible environments

### IoT Industrial Supervision Architecture | *Python, Kafka, Spark, TimescaleDB, MinIO, scikit-learn* 2025

- Ingested real-time sensor streams (temperature, vibration) via Apache Kafka and processed them with Spark Streaming, applying type-aware validation filters before storage
- Persisted raw data to MinIO (S3-compatible) and computed **1-minute sliding window KPIs** into TimescaleDB via JDBC, with **30-second watermarking** for late-arrival tolerance
- Trained a RandomForest model on **1,084 temperature readings** to predict values **5 time steps ahead** using a rolling window of 10 measurements, with live inference fed from Kafka
- Implemented an autonomous safety trigger that automatically shuts down the sensor simulator upon detecting critical temperature thresholds

## TECHNICAL SKILLS

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**Languages:** Python (Pandas, NumPy, Scikit-learn), SQL, TypeScript/JavaScript, HTML/CSS

**Frameworks:** React Native, React JS, Next.js, Django

**Big Data & AI:** Apache Kafka, Apache Spark, Machine Learning, Deep Learning, Generative AI, RAG

**Cloud & Databases:** Azure, PostgreSQL, MySQL, TimescaleDB, MinIO, Supabase, SQLite

**Visualization:** Grafana, Power BI, Apache Superset

**DevOps & Tools:** Docker, Kubernetes, Git/GitHub, Hugging Face, ClearML

EDUCATION

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| <b>EIGSI Casablanca – School of Industrial Systems Engineering</b><br><i>Master’s-level Engineering Degree – Specialization: Artificial Intelligence &amp; Big Data</i> | Casablanca, Morocco<br><i>2022 – 2026</i> |
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CERTIFICATIONS

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| <b>Oracle Cloud Infrastructure 2025 Generative AI Professional</b> – Oracle |
| <b>Oracle Cloud Infrastructure Data Science Professional</b> – Oracle       |

LANGUAGES

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| <b>French:</b> Native (C2)   <b>English:</b> Intermediate (B2) |
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